



PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – MODULE 2

Introduction to Programming

SESSION / ASSIGNMENT – 4

TASKS :

1. Create a simple function called `calculateTotal` that takes two numbers: `itemPrice` and `quantity`, and returns the total bill amount using arithmetic operators.



```
#include <stdio.h>

float calculatetool(float itemprice ,int quantity){

    return itemprice*quantity;
}

int main(){

    float total= calculatetool(599.99,3);

    printf("\nTotal Bill : %.2f \n ",total);

    return 0;
}
```

OUTPUT :

```
Total Bill : 1799.97
```

2. Build a Flipkart-style discount calculator: given product price, discount percentage, and a boolean isMember, use arithmetic and logical operators to calculate the final price (apply an extra 5% off if isMember is true).



```
#include <stdio.h>
#include <stdbool.h>

int main(){

    float product_price=599.99;
    float discount=20.00;
    bool isMember=true;

    float final_price= product_price - (product_price * discount / 100);

    if (isMember=true){
        float final_price= final_price - (final_price * 5 / 100);
    }

    printf("\nFinal Price : %.2f \n ",final_price);

    return 0;
}
```

OUTPUT :

```
Final Price : 479.99
```

3. Write a function `isEligibleForOffer` that takes a user's age and total order value, and returns true if the user is 18 or older AND the order value is above 500, otherwise false.

Hint: Use relational and logical operators together.



```
#include <stdio.h>
#include <stdbool.h>

bool isEligibleForOffer(int age, float order_value){
    return (age >= 18 && order_value > 500);
}

int main(){

if(isEligibleForOffer(20,700)){
    printf("\n True \n ");
}else{
    printf("\n False \n ");
}
return 0;
}
```

OUTPUT:

If the user is 18 or older AND the order value is above 500

True

Else

False

4. Given three variables: likes, comments, and shares (all numbers), write code to check if a post is 'trending' on Instagram (at least 1000 likes OR more than 200 comments AND at least 50 shares). Print the result.



```
#include <stdio.h>

int main(){

    int likes=1150;
    int comments=300;
    int shares=60;

    if(likes >= 1000 || (comments >= 200 && shares >= 50)){
        printf("\n Trending \n ");
    }
    else{
        printf("\n Not Trending \n ");
    }
    return 0;
}
```

OUTPUT:

If (at least 1000 likes OR more than 200 comments AND at least 50 shares)

Trending

Else

Not Trending

5. Write a code snippet that demonstrates the difference between pre-increment (++count) and post-increment (count++) by logging the values before and after using both on a followerCount variable.



```
#include <stdio.h>

int main(){

    int followerCount=10;

    printf("\nPost Increment");
    printf("\nusing post increment %d",followerCount++);
    printf("\nAfter Post Increment %d \n ",followerCount);

    printf("\nPre Increment");
    printf("\nusing Pre Increment : %d",++followerCount);
    printf("\nAfter Pre Increment : %d",followerCount);

    return 0;
}
```

OUTPUT:

```
Post Increment
using post increment 10
After Post Increment 11

Pre Increment
using Pre Increment : 12
After Pre Increment : 12
```